

YI GU

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🎓 EDUCATION

University of California, San Diego Sept. 2022 – Present

Ph.D. student in Data Science, Halıcıoğlu Data Science Institute, anticipated 12/2026

Master of Science in Data Science (earned en route)

Advisor: Prof. Zhiting Hu

Shanghai Jiao Tong University Sept. 2018 – June 2022

Bachelor of Engineering in Computer Science and Technology

- Member of ACM Class, an elite CS program for top 5% talented students.
- GPA: All 3.94 / 4.3, ranked 4 / 32; Core 3.96 / 4.3, ranked 2 / 32.

🔍 RESEARCH INTERESTS

My research interest lies in **foundational models** and **reasonings**. I currently working on **building World Models** (PAN) and related reasoning systems, aiming for the next-generation machine reasoning beyond LLMs. I focus more on the **ML-infra** starting from PAN project, including CUDA kernels, distributed systems, compiles and graphs, and detailed profiling, for both training and inference. Previously, I work on the **training algorithm** and **reasoning ability** of LLMs (RAP, Reasoners) and world models (Pandora).

👥 EXPERIENCES

MBZUAI Institution of Foundation Model (IFM) Sept. 2025 - Present

Research Scientist / Research Intern

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) June 2023 – Sept. 2023

Visiting Student Advisor: Prof. Eric P. Xing

MIT-IBM Watson AI Lab Jan. 2021 – Jan. 2022

Remote Research Intern Advisor: Dr. Chuang Gan and Prof. Joshua B. Tenenbaum

📄 PUBLICATIONS

Selected Publications

PAN: A world model for general, interactable, and long-horizon world simulation

PAN Team Institute of Foundation Models: Jiannan Xiang, **Yi Gu**, Zihan Liu, Zeyu Feng, Qiyue Gao, Yiyan Hu, Benhao Huang, Guangyi Liu, Yichi Yang, Kun Zhou, Davit Abrahamyan, Arif Ahmad, Ganesh Bannur, Junrong Chen, Kimi Chen, Mingkai Deng, Ruobing Han, Xinqi Huang, Haoqiang Kang, Zheqi Liu, Enze Ma, Hector Ren, Yashowardhan Shinde, Rohan Shingre, Ramsundar Tanikella, Kaiming Tao, Dequan Yang, Xinle Yu, Cong Zeng, Binglin Zhou, Zhengzhong Liu, Zhiting Hu, Eric P. Xing

Preprint

Accelerating Structured Chain-of-Thought in Autonomous Vehicles

Yi Gu, Yan Wang, Yuxiao Chen, Yurong You, Wenjie Luo, Yue Wang, Wenhao Ding, Boyi Li, Heng Yang, Boris Ivanovic, Marco Pavone

Preprint

Efficient Long-context Language Model Training by Core Attention Disaggregation

Yonghao Zhuang, Junda Chen, Bo Pang, **Yi Gu**, Yibo Zhu, Yimin Jiang, Ion Stoica, Eric Xing, Hao Zhang

MLSys 2026

Reasoning with Language Model is Planning with World Model

Shibo Hao*, **Yi Gu***, Haodi Ma, Joshua Jiahua Hong, Zhen Wang, Daisy Zhe Wang, Zhiting Hu

EMNLP 2023

LLM Reasoners: New Evaluation, Library, and Analysis of Step-by-Step Reasoning with Large Language Models

Shibo Hao*, **Yi Gu***, Haotian Luo*, Tianyang Liu, Xiyan Shao, Xinyuan Wang, Shuhua Xie, Haodi Ma, Adithya Samavedhi, Qiyue Gao, Zhen Wang, Zhiting Hu

COLM 2024

LLM360 K2-65B: Scaling Up Fully Transparent Open-Source LLMs

Zhengzhong Liu, Bowen Tan, Hongyi Wang, Willie Neiswanger, Tianhua Tao, Haonan Li, Fajri Koto, Yuqi Wang, Suqi Sun, Omkar Pangarkar, Richard Fan, **Yi Gu**, Victor Miller, Liqun Ma, Liping Tang, Nikhil Ranjan, Yonghao Zhuang, Guowei He, Renxi Wang, Mingkai Deng, Robin Algayres, Yuanzhi Li, Zhiqiang Shen, Preslav Nakov, Eric Xing

Preprint

LLM360: Towards Fully Transparent Open-Source LLMs

Zhengzhong Liu, Aurick Qiao, Willie Neiswanger, Hongyi Wang, Bowen Tan, Tianhua Tao, Junbo Li, Yuqi Wang, Suqi Sun, Omkar Pangarkar, Richard Fan, **Yi Gu**, Victor Miller, Yonghao Zhuang, Guowei He, Haonan Li, Fajri Koto, Liping Tang, Nikhil Ranjan, Zhiqiang Shen, Xuguang Ren, Roberto Iriondo, Cun Mu, Zhiting Hu, Mark Schulze, Preslav Nakov, Tim Baldwin, Eric P. Xing

COLM 2024

Other Selected Publications

Pandora: Towards General World Model with Natural Language Actions and Video States

Jiannan Xiang*, Guangyi Liu*, **Yi Gu***, Qiyue Gao, Yuting Ning, Yuheng Zha, Zeyu Feng, Tianhua Tao, Shibo Hao, Yemin Shi, Zhengzhong Liu, Eric P. Xing, Zhiting Hu

Preprint

Text Grafting: Near-Distribution Weak Supervision for Minority Classes in Text Classification

Letian Peng, **Yi Gu**, Chengyu Dong, Zihan Wang, Jingbo Shang

EMNLP 2024

Language Models Meet World Models: Embodied Experiences Enhance Language Models

Jiannan Xiang*, Tianhua Tao*, **Yi Gu**, Tianmin Shu, Zirui Wang, Zichao Yang, Zhiting Hu

NeurIPS 2023

Revisiting the Roles of “Text” in Text Games

Yi Gu*, Shunyu Yao*, Chuang Gan, Joshua B. Tenenbaum, Mo Yu

Findings of EMNLP 2022

Finding Fallen Objects Via Asynchronous Audio-Visual Integration

Chuang Gan*, **Yi Gu***, Siyuan Zhou, Jeremy Schwartz, Seth Alter, James Traer, Dan Gutfreund, Joshua Tenenbaum, Josh McDermott and Antonio Torralba

CVPR 2022

🏆 HONORS AND AWARDS

Champion, ICPC 2018 Singapore Site	Dec. 2018
Gold Medal, ICPC 2019 Xuzhou Site, ranked 4th	Nov. 2019
National Scholarship of P.R. China (Top 0.2% nationwide), Ministry of Education of P.R. China	2021
Zhiyuan Outstanding Student Scholarship, SJTU	2022
Rong Chang Scholarship in Technological Innovation (2 times) (20 winners each year), SJTU	2019, 2020
Zhiyuan Honorary Scholarship (3 times)	2019, 2020, 2021

⚙️ SKILLS

Programming Languages: Python = Triton = Kotlin = Java > C/C++ = HTML = Javascript

Human Languages: English - Fluent, Chinese - Native speaker, Japanese - Conversational

Tech: ML Infra (CUDA, PyTorch, Distributed, ...) = ML Algorithms (LLM, DiT, ...) ≫ Frontend (Android, Web) > (what coding agents can do)